

Maxence Caucheteux

Curriculum Vitae

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🐙 Github in LinkedIn

Education

- 2023 – 2026 **École Nationale des Ponts et Chaussées**, *Engineering Degree*, Paris.
Relevant coursework: Stochastic Processes, Stochastic Calculus, Advanced Statistics, Partial Differential Equations, Optimisation, Control Theory, Functional Analysis, Programming, Machine Learning
- 2025 – 2026 **École Polytechnique – Sorbonne Université**, *MSc Probability and Finance (El Karoui Master)*, Paris.
Relevant coursework: Stochastic Modeling of Derivatives (E. Abi Jaber), High Frequency Trading (M. Rosenbaum), Rate Models (N. El Karoui), Stochastic Calculus (N. Broutin), Numerical Methods (G. Pagès), Stochastic Control (I. Kharroubi), Machine Learning and Deep Learning (P. Gallinari), Risk Measures (A. Alfonsi)
- 2021 – 2023 **Lycée Louis-le-Grand**, *Preparatory Classes MP2I – MPI**, Paris.
Intensive preparation for the nationwide competitive entrance exams to French Grandes Écoles.
Relevant coursework: Mathematics, Physics, Computer Science, Philosophy, English

Work Experience

- 2026 **Quantitative Researcher Intern**, *Barclays*, London, United Kingdom.
Six-month internship within the Cross-Market Quantitative Analytics team. Topic: Assessing the impact of stochastic volatility on XVA calculations.
- 2025 **Assistant Lecturer in Mathematics**, *University of Paris*, Paris.
Taught mathematics tutorials to approximately 40 second-year management students as part of the Quantitative Techniques for Economics course. Designed exam problems and graded examinations on several occasions.

Academic Projects

- 2025 **State spaces of multifactor approximations of Volterra processes.**
Research internship supervised by Aurélien Alfonsi. Study of the state spaces of multifactor Markovian approximations of nonnegative Volterra processes, extending results from the literature (Abi Jaber).
- 2025 **Currency option pricing and hedging under the Garman-Kohlhagen model.**
Project on FX options including pricing and delta computation for European options, delta hedging simulations, and numerical solution of the pricing PDE for American options using the Brennan-Schwarz algorithm.
- 2025 **Calibration methods for financial risk models.**
Project proposed by Milliman comparing maximum likelihood estimation and generalized method of moments for calibrating financial risk models.
- 2024 **Mathematical modelling of epidemic propagation.**
Project with Jean-François Delmas studying the influence of population mobility on the spread of epidemics using stochastic and deterministic models.
- 2023 **Classification of integer simplices.**
Research project in discrete geometry aimed at classifying integer simplices according to their number of lattice points and developing algorithms to compute this classification.

Skills and Interests

Programming Python, C/C++

Languages French (Native), English (Advanced)

Interests Running (5 times per week, 5km PB: 17:34, 10km PB: 37:16)

Referees

Aurélien Alfonsi

*Professor of Stochastic Calculus and Mathematical
Finance*

CERMICS, École des Ponts et Chaussées

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